New Constraints on the Stellar Mass-Halo Mass Relation: Stacked Lensing of Dwarf Galaxies

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Abstract

The stellar-to-halo mass relation (SHMR) is a powerful tool for estimating the dark matter content of a galaxy from its observed stellar mass. The SHMR at the dwarf galaxy scale (halo mass,$M_h = 10^{10} - 10^{11}\,\mathrm{M}_{\odot}$) is particularly informative because it links galaxy formation to small-scale dark-matter structure, where different galaxy evolution models can diverge most strongly. However, at the low-mass end, many SHMR constraints rely on indirect halo-mass inference, leaving systematic uncertainties from selection effects, baryonic physics, and environment that are not calibrated by an independent mass probe. We propose to independently calibrate the dark matter halo masses of dwarf galaxies at low redshift ($z \sim 0.2 - 0.8$) by stacking lensing shear around large samples of dwarfs ($\sim 10^5$) in the cutting-edge Roman High-Latitude Survey data to boost the signal-to-noise ratio. This program will produce the first statistically significant sample of dwarf galaxies with well-defined halo masses, which is critically needed to reduce the scatter in the low-mass SHMR by $50 - 70\%$. Establishing an independent halo-mass scale for low-mass galaxies will shed new light on the galaxy–halo connection and help us better understand how galaxies' stellar masses and their dark matter halos grow over cosmic time.

■ Scientific Justification

The Galaxy-Halo Connection at the Dwarf Scale

The stellar mass – halo mass relation (SHMR) links galaxy formation physics to dark matter structure. The SHMR at the dwarf galaxy scale (halo mass,$M_h = 10^{10} - 10^{11}$ M$_\odot$) is especially important because it probes the regime where galaxy-formation physics and small-scale dark-matter models diverge most strongly. Constraining the SHMR below the Milky Way scale is therefore critical for distinguishing among different galaxy formation models (Wang et al. 2026). The divergence among models arises from multiple physical processes: different quenching prescriptions, such as the timing of quenching, can shift the typical halo mass at fixed stellar mass by ~ 0.15 dex (O'Leary et al. 2023), while variations in merger history further modify stellar mass growth and thus the mapping between stellar and halo mass (Deason et al. 2022). **A well-measured dwarf-galaxy SHMR is therefore essential for constraining the physics of galaxy evolution** (e.g., quenching and merger histories) and for testing the galaxy–halo connection in the low-mass regime.

However, we cannot test these competing models because low-mass SHMR constraints rely on indirect halo-mass inference, leaving systematic uncertainties from selection effects, baryonic physics, and environment that are not calibrated by an independent mass probe. The primary inference methods, such as abundance matching, are sensitive to incompleteness in the faint end of the stellar mass function; while H I kinematics is limited to nearby, gas-rich systems (Wang et al. 2026). Solving this problem is *urgent now* because Roman, Euclid, and Rubin will soon enable precision studies of large dwarf-galaxy samples, and those analyses will otherwise inherit the same poorly calibrated systematics.

We propose to directly calibrate the dark matter halo masses of dwarf galaxies at $z \sim 0.2 - 0.8$ by stacking weak-lensing shear around a sample of $\sim 10^5$ photometrically selected dwarfs in the Roman High-Latitude Wide-Area Survey (HLWAS). Roman's primary science driver is weak-lensing cosmology. Its wide-field infrared imager (WFI) provides the combination of wide survey area (~ 2000 deg^2), depth ($J \sim 26.7$), and shape-measurement quality needed to detect the lensing signal around dwarf lenses. This is a measurement infeasible with current ground-based surveys. Our strategy is to identify $\sim 10^5$ dwarf galaxies from photometric stellar masses and redshifts, measure the shapes of background galaxies, and stack the tangential shear signal in stellar-mass bins to recover the corresponding halo masses; slitless spectroscopy will provide a complementary kinematic-lensing cross-check where available. This program will deliver **the first independently calibrated SHMR at dwarf galaxy mass scale**, and a reusable Roman stacked-lensing pipeline, providing a lasting resource for studies of galaxy evolution and dark matter physics.

The Need for Direct Dwarf Halo-Mass Measurements

The dwarf SHMR is key to understanding the galaxy-halo connection and low-mass galaxy evolution. However, inferring dwarf dark matter halo masses indirectly through their baryonic counterparts introduces systematic biases and uncertainties into the SHMR. For instance, tidal stripping can remove dark matter, biasing halo mass estimates (Fattahi et al. 2018), while stellar feedback can reshape the inner dark-matter profile, altering observed velocity profiles even at fixed dark-matter mass (Chan et al. 2015). Furthermore, low-surface-brightness dwarfs may be entirely missing from observational samples, introducing severe selection bias (Blanton et al. 2005; Greco et al. 2018). **Thus, directly measuring dwarf galaxy halo masses is essential to securely anchor the low-mass SHMR**, ultimately allowing us to distinguish among competing models of galaxy and dark matter halo evolution.

We currently lack a *direct, model-independent halo-mass measurement* for a statistically large sample of dwarf galaxies to bypass the biases and break the degeneracy between competing galaxy evolution models. This program will produce the **first statistically significant sample of $\sim 10^5$ dwarf galaxies with well-defined halo masses**, critically needed to solve this problem.

This Proposal: Stacked Weak Lensing of Dwarf Galaxies

In this proposal, we will measure and stack the weak-lensing shear signal from background galaxies around dwarf galaxies at $z \sim 0.2 - 0.8$ in the Roman HLWAS. Roman's WFI provides the *unique combination* of wide area ($\sim$ 2000 deg^2), infrared depth ($J \sim 26.7$), and space-based point spread function (PSF) stability required for precision shape measurements around faint, low-mass lens galaxies. **No ground-based survey achieves the simultaneous area, depth, and shape-measurement quality necessary for this analysis.**

We propose utilizing the Roman WFI to observe the weak-lensing signal around $\sim 10^5$ dwarf galaxies within the Roman HLWAS. Target galaxies will be identified using photometric stellar masses ($M_\star \sim 10^7 - 10^9\,\mathrm{M}_\odot$) derived from Roman multi-band imaging and photometric redshifts. Subsequently, we will measure the shapes of background source galaxies using WFI broad-band imaging, leveraging the stable, space-based PSF for high-fidelity shape calibration. Where available, slitless grism spectroscopy will provide a complementary kinematic-lensing cross-check. As shown in Figure 1, stacking the tangential shear signal in stellar-mass bins will yield the S/N required to constrain dwarf halo masses and directly test competing SHMR models.

This program will deliver **the first independently calibrated SHMR at the dwarf galaxy mass scale**, reducing scatter by $\sim 50 - 70\%$ relative to indirect constraints. By providing a *critical ground truth* for validating abundance matching and kinematic methods, this independent calibration secures the unbiased mass mapping required to break the degeneracy among competing galaxy evolution models.

A Reusable Legacy Framework for the Roman Observatory

The independently calibrated SHMR produced by this program provides the critical ground truth needed to test whether foundational indirect methods, like abundance matching, carry systematic biases. By securely calibrating these indirect mass proxies, this program advances the subfield by delivering the empirical anchor needed to distinguish among galaxy formation models that currently diverge at the dwarf scale.

As the community prepares to use Roman, Euclid, and Rubin data to constrain cosmology and galaxy evolution, an independent halo-mass calibration at the dwarf scale is urgently needed to prevent the propagation of systematic biases from indirect methods into next-generation analyses. This proposal directly addresses two of Roman's key science themes: "Dark Matter" (constraining halo masses and small-scale structure) and "Cosmic Evolution" (understanding how galaxies and halos co-evolve). This study will increase the legacy value of the Roman Observatory by providing a **public dwarf halo-mass catalog** and a reusable stacked-lensing pipeline that will serve the community at large, enabling diverse galaxy evolution studies and contributing to Roman's cosmological weak-lensing objectives.

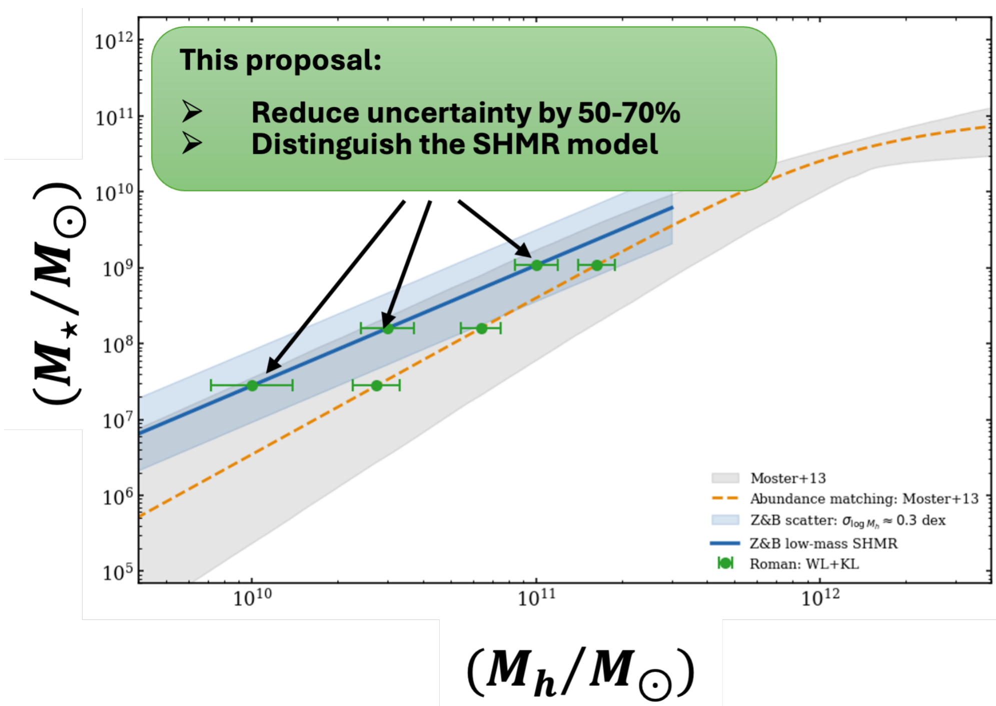

Figure 1: Roman stacked lensing will provide the first direct, model-independent calibration of the dwarf SHMR, resolving current model degeneracies. The blue solid line and shaded region show the SHMR from Zaritsky & Behroozi (2023)—photometrically calibrated against dynamical mass estimates—with its 3σ scatter. The orange dashed line and grey band represent the abundance-matched relation from Moster et al. (2013) and its associated 3σ scatter. Current SHMR constraints diverge significantly below $M_h \sim 10^{11}\,\mathrm{M}_\odot$ because they rely on indirect mass proxies and assumed dark-matter profiles. <u>This program will utilize $\sim 10^5$ HLWAS dwarfs to provide an absolute halo-mass anchor (green points) via weak-lensing and kinematic-lensing (WL+KL) constraints, yielding the ground truth required to distinguish between competing galaxy formation and dark matter models.</u>

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